

VR Temple Run

Swiping away Temple Run was quite amazing on a smartphone, but will it be exciting on a virtual headset? <http://dgit.in/VirtTempRun>



Security upgraded

Following the series data breaches worldwide, South Korea has boosted its cyber security in nuclear plants <http://dgit.in/SecNucReactor>

Industry Connect

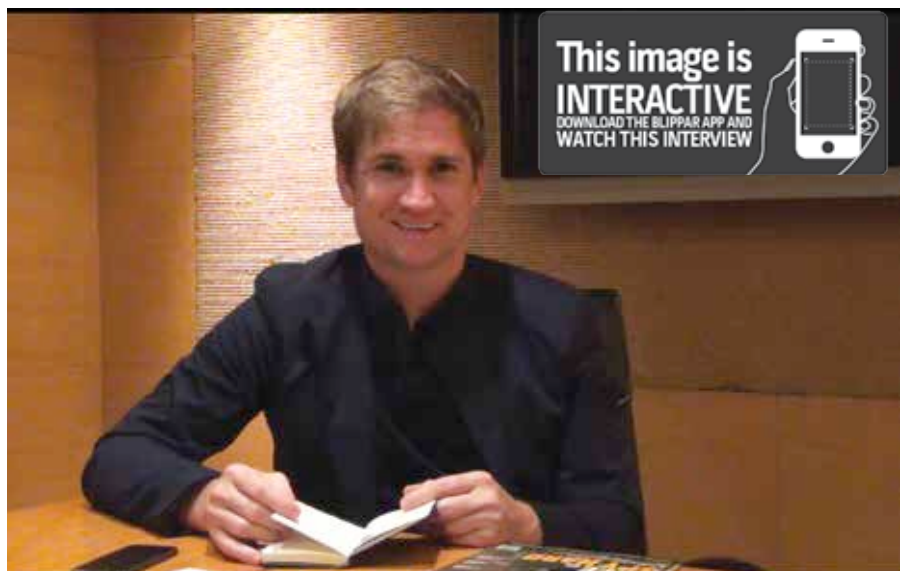
most people don't have the time to deal with. So the hardware needs to evolve.

What exactly is 4D Printing?

Jordan: The best analogy to 4D Printing is in biological systems. Essentially we ourselves are 4D printers right? We are built additively and we modify ourselves over time, so the precedent is biology and the living world. 4D printing in the modern sense is being able to additively manufacture components that can assemble themselves or modify themselves over time. You don't have static products now. Think of products that can move and adapt, not just in terms of their software interface but in terms of their physical interface.

Would the analogy go further? Would it be ok to think that such products would multiply as well as mould themselves as cellular organisms do?

Jordan: Let's say that they will at least modify themselves. In a certain case they may multiply if the need arises. Imagine that this table was a 4D printed object and it was made of smaller components and imagine suddenly you had a need in this room to have not one large table but five smaller tables then they could reprogram themselves to be five smaller tables, so in that sense, yes, potentially they can



objects that look amazing. Often times they look like something you've never seen before. So on the consumer side, I think technology is coming now, we have been restricted largely to fused deposition modelling and extrusion technologies and that will advance a lot, I mean there's post-processing and things that you can do to make it look better, to make it feel like a finished object, but there's always ways to embrace the fact that it is 3D printed and maybe this is a new aesthetic of objects that we should embrace. I

UV light and it's continuously extruding a form out of a liquid and the results are very high quality. It doesn't have the readable layers you typically see with FDM, you get very high precision, and you can make very small artifacts with it. We are actually printing hair with it. It feels like fur, that's the kind of resolution that we can get. So as new technologies come online, some of the perception of 3D printing will change.

So do you foresee anything like the Star Trek food synthesizer or things like that happening any time soon?

Jordan: It will happen, it is going to be in some time, because it has some chemical issues that we have to overcome. At the end of the day you will have to get molecules to bind and so within a single material system, we're going to have a lot of advances very quickly, in fact there some very compelling work going on right now around the world in metals printing, ceramics, polymers and mixing different materials within a family. But trying to get metals and plastics to chemically bond, you know one is using ionic bonding mechanisms, one is using covalent bonding mechanisms, that's fundamental physics, and we can't change that. So we need to come up with some clever ways to be able to mix material systems to be able to print as the replicator in Star Trek does, it will happen but it's going to be quite a ways down the road. **1**

The F-18 has had laser-sintered parts, I think, for the last 12 years, commercial aircrafts now have 3D printed parts as well, and if you go into factories, you see 3D Printing making its way into making fixtures, so that humans are more efficient

multiply. But the material has to obviously come from somewhere, so we are restricted by the laws of physics.

Right now the appeal of 3D printing is only restricted towards faster prototyping of products isn't it? 3D printed objects still don't look as attractive as finished goods. At least the ones we make as hobbyists, so when do you think that would change?

Jordan: On the consumer side I agree with you. On the industrial side 3D printed

mean the worst thing that you could do is try to take a new technology and try to do what you did before with it. It would be like taking your mobile phone and plugging it into your landline and saying well it doesn't work very well. So I think there would be some change in perception and adoption of the products being made by 3D printing because it won't look like what we had before. The new Autodesk printer Ember is using a DLP technology, which is a liquid resin, it's exposing it to